

1. Consider an MDP having three states S_1 , S_2 , and S_3 , with rewards -1 , -2 , and 0 for executing an action from that state, respectively. S_3 is the terminal state where only exit action is available that gives no reward. There is no other living reward. S_1 and S_2 each have two possible actions: a and b . 12
(CO1)
(PO1)

The transition function is described as follows:

- In S_1 , action a moves the agent to S_2 with probability 0.8 and makes the agent stay put with probability 0.2 .
- In S_2 , action a moves the agent to S_1 with probability 0.8 and makes the agent stay put with probability 0.2 .
- In either S_1 or S_2 , action b moves the agent to S_3 with probability 0.1 and makes the agent stay put with probability 0.9 .

Assume that the discount factor, $\gamma = 1$ and the initial policy has action a in both states.

With detailed steps, recommend the optimal policy for each state via policy iteration.

Solution:

Initially, $\pi(S) = a$ for $S \in \{S_1, S_2\}$. $R(S_1) = -1$, $R(S_2) = -2$, and $R(S_3) = 0$.

Policy Evaluation:

$$V(S_1) = -1 + 0.2 \times V(S_1) + 0.8 \times V(S_2)$$

$$V(S_2) = -2 + 0.8 \times V(S_1) + 0.2 \times V(S_2)$$

$$V(S_3) = 0$$

We notice that $V(S_1)$ and $V(S_2)$ are inconsistent. If we were to try to solve them iteratively, we would find the values tending to $-\infty$. So this problem is unsolvable.

Rubric:

- 10 points for identifying that it goes to negative infinity.
- 2 points for decision.

2. A meme is an image, video, piece of text, etc., typically humorous in nature, that is copied and spread rapidly by internet users, often with slight variations. 3

Create a meme related to this course. A text description will suffice.

Solution:

**ARTIFICIAL INTELLIGENCE
WHEN NATURAL STUPIDITY WALKS IN**



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